

AI & ML Task 4 – Mandatory Deliverables Report

Classification Models, Evaluation Metrics & Handling Imbalanced Data

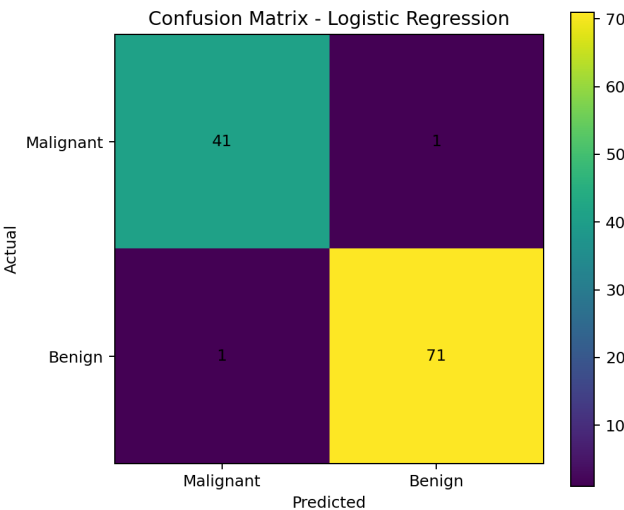
Dataset: Breast Cancer Dataset from scikit-learn

A binary classification system was developed using Logistic Regression, Balanced Logistic Regression and Decision Tree. The workflow follows the Maincrafts Technology Task 4 requirements.

1. Dataset and Preprocessing

The dataset contains 569 observations and 30 numerical features. The target classes are malignant (0) and benign (1). An 80:20 stratified train-test split was used with random_state=42. StandardScaler was fitted only on the training data and then applied to the test data to prevent data leakage.

2. Confusion Matrix



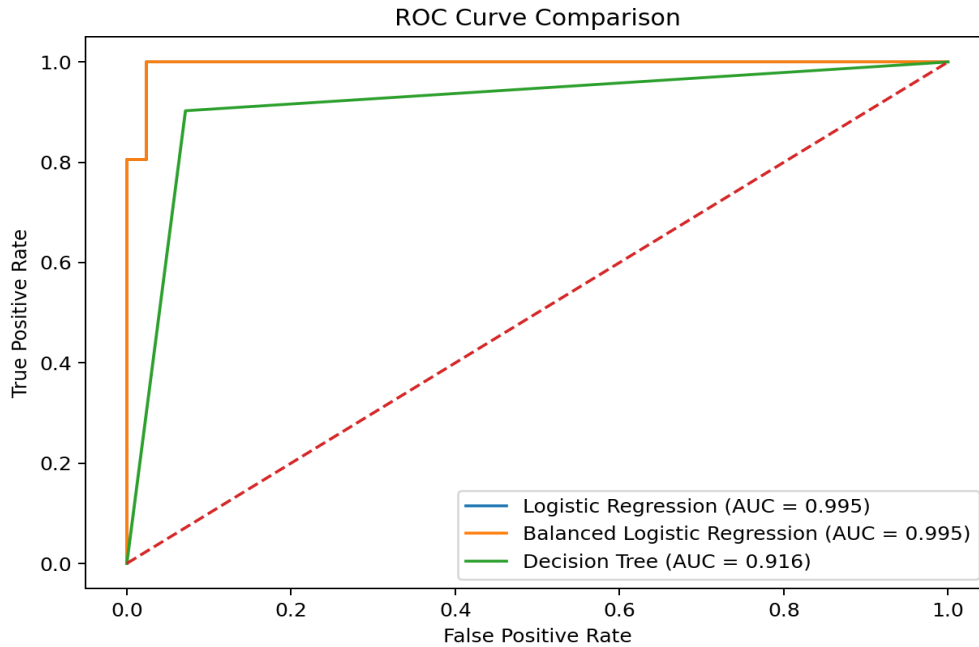
The confusion matrix provides detailed error analysis through true positives, true negatives, false positives and false negatives. In medical diagnosis, false negatives are especially important because they represent missed positive cases.

3. Evaluation Metrics

Accuracy measures overall correctness. Precision measures the correctness of positive predictions. Recall measures the proportion of actual positive cases detected. F1-score balances precision and recall. ROC-AUC measures the model's ability to distinguish between the two classes across thresholds.

Baseline Logistic Regression achieved Accuracy = 0.9825, Precision = 0.9861, Recall = 0.9861, F1-score = 0.9861, and AUC = 0.9954.

4. ROC Curve and AUC



The ROC curve plots the true positive rate against the false positive rate at different thresholds. An AUC closer to 1 indicates stronger discrimination between the two classes.

5. Handling Class Imbalance

Class imbalance can make accuracy misleading because a classifier may favor the majority class. To address this, Balanced Logistic Regression uses `class_weight='balanced'`, which automatically increases the importance of the less frequent class during training.

6. Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
Balanced Logistic Regression	0.9561	0.9855	0.9444	0.9645	0.9954
Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157

7. Final Model Decision

The final model should not be selected using accuracy alone. For medical diagnosis, recall is particularly important because a false negative can mean that a malignant case is missed. F1-score and ROC-AUC provide additional evidence, while interpretability and stability should also be considered.